RESUME

Dr. Chandrashekhar Singh

Ph.D. (Biotechnology)

Institute of Science, Department of Botany

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Research interests

- Plant Biotechnology & Clinical Microbiology
- Biological Evaluation of Herbal and Nanomedicines
- Development of Novel Bioinspired Agents for Wound Healing

Teaching Experience:

1. **Assistant Professor**
   Maharishi University of Information Technology, Lucknow
   August 1, 2024 – December 30, 2024
2. **Guest Faculty**
   Sri Agrasen Kanya PG College, Bulanala, Varanasi
   January 24, 2019 – March 18, 2020

Research Experience

- **Postdoctoral Researcher (ICMR/RA)**
  Department of Pharmaceutical Engineering and Technology, IIT BHU, Varanasi, Uttar Pradesh, (February 8, 2021 – January 7, 2024)
- **Senior Fellow**
  Department of Crop Improvement, Indian Institute of Vegetable Research, Varanasi (Sept 2012 – March 2013)

Educational Qualification

Ph.D., **(2018) (Biotechnology)** Banaras Hindu University (Institute of Science), Department of Botany.

Supervisor: Professor K.N. Tiwari

Title of Ph.D.: Pharmacognostical, phytochemical and Bioactivity Studies of *Premna integrifolia*.

M.Sc., (Biotechnology) Bundelkhand University, Jhansi, India, 2009

Research Project: Cloning and expression of MCE-4A gene and expression of *mce4a* protein.

Project Advisor: Professor Mridula Bose (University of Delhi)

B.Sc., (Zoology and Chemistry), University of Allahabad, Allahabad, India, 2005

Scholarships and Awards

- PhD Fellowship, at Banaras Hindu University from University Grant Commission (UGC), New Delhi, India.
- Qualified Junior Research Fellowship and National Eligibility Test for the lectureship from CSIR (2011).

Publications

1. Verma, Pooja, Jyoti Dixit, **Chandrashekhar Singh**, Alakh Narayan Singh, Aprajita Singh, Kavindra Nath Tiwari, Madaswamy S. Muthu, Gopal Nath, and Sunil Kumar Mishra. "Preparation of hydrogel from the hydroalcoholic root extract of *Premna integrifolia* L. and its mediated green synthesis of silver nanoparticles for wound healing efficacy." *Materials Today Communications* (**2024**): 110228. (**IF:3.7**)
2. **Chandrashekhar Singh,** Abhishesh Kumar Mehta, Matte Kasi Vishwanadh, Punit Tiwari, Rajesh Saini, Sanjeev Kumar Singh, Ragini Tilak, Kavindra Nath Tiwari, Madaswamy S. Muthu "Chitosan film of thiolated TPGS-modofied Au-Ag nanoparticles for combating multidrug- resistant bacteria" Colloids and Surfaces A: Physicochemical and Engineering Aspects, **2024**, 686, 133287. (**IF: 5.2)**.
3. Ankit Malik, **Chandrashekhar Singh** "Nanofibers of N, N, N-trimethyl Chitosan Capped Bimetallic Nanoparticles: Preparation, characterization, Wound dressing, and In vivo treatment of MDR microbial infection and Tracking by Optical and Photoacoustic Imaging "International Journal of Biological Macromolecules. **2024,** 263, 130-154. **(IF:8.2).**
4. Upadhyay, Richa, **Chandrashekhar Singh**, Awadhesh Kumar Mishra, Rajesh Saini, Jasmeet Singh, and Kavindra Nath Tiwari. "Estimation of Antioxidant Potential, Phytochemical Profiling, and In Silico Characterization of Hepatoprotective Biomarkers of *Phyllanthus fraternus* GL Webster Leaves." *Chemistry Select* 8, no. 19 (**2023**): e202300581. (**IF: 2.3)**
5. **Chandrashekhar Singh**, Sumit Kumar Anand, Richa Upadhyay, Nidhi Pandey, Pradeep Kumar, Deepjyoti Singh , Punit Tiwari , Rajesh Saini , Kavindra Nath Tiwari , Sunil Kumar Mishra , Ragini Tilak. Green synthesis of silver nanoparticles by root extract of *Premna integrifolia* L. and evaluation of its cytotoxic and antibacterial activity. Materials Chemistry and Physics, 297, 1, (**2023**), 127413. **IF:4.8)**
6. **Singh, Chandrashekhar**, Abhishesh Kumar Mehata, Vikas, Punit Tiwari, Aseem Setia, Ankit Kumar Malik, Sanjeev K Singh, Ragini Tilak and Madaswamy S Muthu. Design of Novel Bioadhesive chitosan film loaded with bimettalic gold-silver nanoparticles for antibiofilm and wound healing activity. Biomedical Materials, (**2023**) DOI 10.1088/1748-605X/acb89b. (**IF:4.1)**
7. **Singh, Chandrashekhar**, Abhishesh Kumar Mehata, Vishnu Priya, Ankit Kumar Malik, Aseem Setia, M. Nikitha Lakshmi Suseela, Patharaj Gokul, Sanjeev K. Singh, and Madaswamy S. Muthu. "Bimetallic Au–Ag Nanoparticles: Advanced Nanotechnology for Tackling Antimicrobial Resistance." *Molecules* 27, no. 20 (**2022**): 7059.**IF:(4.6)**
8. Mehata, Abhishesh Kumar, M. Nikitha Lakshmi Suseela, Patharaj Gokul, Ankit Kumar Malik, Matte Kasi Viswanadh, **Chandrashekhar Singh**, Joseph Selvin, and Madaswamy S. Muthu. "Fast and highly efficient

liquid chromatographic methods for qualification and quantification of antibiotic residues from environmental waste." Microchemical Journal (**2022**): 179,107573. **IF:5)**

9. **Chandrashekhar Singh**, Kavindra Nath Tiwari, Pradeep Kumar, Anil Kumar, Jyoti Dixit, Rajesh Saini, Sunil Kumar Mishra, "Toxicity profiling and antioxidant activity of ethyl acetate extract of leaves of *Premna integrifolia* L. for its application as protective agent against xenobiotics" Toxicology Reports, 8, **2021**, 196-205.
10. **Singh C**, Upadhyay R, Tiwari KN. Comparative analysis of the seasonal influence on polyphenolic content, antioxidant capacity, identification of bioactive constituents and hepatoprotective biomarkers by in silico docking analysis in *Premna integrifolia* L. Physiol Mol Biol Plants. 2022.(**IF:3.1)**
11. **Singh, Chandrashekhar**, Chandra Prakash, Pallavi Mishra, Kavindra Nath Tiwari, Sunil Kumar Mishra, Raghunath Shahaji More, Vijay Kumar, and Jasmeet Singh. "Hepatoprotective efficacy of *Premna integrifolia* L. leaves against aflatoxin B1-induced toxicity in mice." Toxicon (2019) 166: 88-100. (**IF:3.23)**
12. **Singh, Chandrashekhar**, Jitendra Kumar, Pradeep Kumar, Brijesh Singh Chauhan, Kavindra Nath Tiwari, Sunil Kumar Mishra, S. Srikrishna, Rajesh Saini, Gopal Nath, and Jasmeet Singh. "Green synthesis of silver nanoparticles using aqueous leaf extract of *Premna integrifolia* (L.) rich in polyphenols and evaluation of their antioxidant, antibacterial and cytotoxic activity." Biotechnology & Biotechnological Equipment (2019):33: 1-13. **IF:(1.42)**
13. **Singh Chandrashekhar**, Prakash Chandra, Tiwari Kavindra Nath, Mishra Sunil Kumar, Kumar Vijay, *Premna integrifolia* ameliorates cyclophosphamide-induced hepatotoxicity by modulation of oxidative stress and apoptosis, Biomedicine & Pharmacotherapy, 2018.107, 634-643. (**IF: 7.6)**
14. Richa Upadhyay, Sarvesh Pratap Kashyap**, Chandrashekhar Singh**, Kavindra Nath Tiwari, Karuna Singh, Major Singh, Ex situ conservation of *Phyllanthus fraternus* Webster and evaluation of genetic fidelity in regenerates using DNA-based molecular marker. Applied Biochemistry and Biotechnology.2014. 174 (6), 2195-2208. (**IF:2.2)**
15. Richa Upadhyay, Sarvesh Kashyap, **Chandrashekhar Singh**, Kavindra Nath Tiwari, Karuna Singh, Major Singh, Assessment of factors on shoot proliferation potential of nodal explants of *Phyllanthus fraternus* and assessment of genetic fidelity of micropropagated plants using RSSR primer.Biologia.2014.69.1685-1692. **IF:(0.8)**
16. Pandey, L.K., Ojha, K.K., Singh, PK., **Singh, C.S.,** Dwivedi, S., Bergey, E.A. (2016). Diatoms image database of India: a research tool. Environmental Technology and Innovation. 5,148-160. **IF: (7.6)**
17. Sunil Kumar Mishra, Bhoosan Yadav, Prabhat Upadhyay, Pradeep Kumar, **Chandrashekhar Singh**, Jyoti Dixit and Kavindra Nath Tiwari. (2018). LC-ESI MS/MS Profiling, Antioxidant and Anti-Epileptic Activity of *Luffa cylindrica* (L.) Roem Extract. Journal of Pharmacology and Toxicology. ISSN 1816-496XDOI: 10.3923/jpt.2018.
18. **Chandrashekhar Singh**, Sumit Kumar Anand, Kavindra Nath Tiwari, Sunil Kumar Mishra, Poonam Kakkar, Phytochemical profiling and cytotoxic evaluation of *Premna serratifolia* L. against human liver cancer cell line. 3 Biotech (2021) 11:115. **IF:(3.02)**

19. Singh, Chandrashekhar, Anjali Singh, Deepjyoti Singh, and Richa Upadhyay. "Potential therapeutic solution for *Clostridioides difficile* infection: current scenario and future prospects." *Medicine in Microecology* (2025): 100121.

Book Chapter:

1.Chandrashekhar Singh, Rajesh Saini, Richa Upadhyay and Kavindra Nath Tiwari (**2023**). Revealing the epigenetic Mechanisms Underlying the stress responses in Medicinal plants. Stress responsive factors and Molecular farming in Medicinal Plants, Springer. (Edited by Divya Singh, Amit Kumar Mishra, and Akhileshwar Kumar Shrivastav). https://doi.org/10.1007/978-981-99-4480-4

2.Vikas, Mehata, A.K., **Singh, C.,** Malik, A.K., Setia, A., Muthu, M.S. (2023). Alginate in Cancer Therapy. In: Jana, S., Jana, S. (eds) Alginate Biomaterial. Springer, Singapore. https://doi.org/10.1007/978-981-19-6937-9_11

3. Kavindra Nath Tiwari, Vaibhav Tiwari, Awadhesh Kumar Mishra, **Chandrashekhar Singh** and Richa Upadhyay, Germplasm conservation of *Bacopa monnieri* Wettst. under slow growth conditions by encapsulation of nodal explants. Assessment and conservation of forest genetic resources through biotechnological interventions. (Edited by Sanjay Singh and Rameshwar Das).

4. Kavindra Nath Tiwari, Awadhesh Kumar Mishra, **Chandrashekhar Singh** and Jasmit Singh, Biodiversity of medicinal plants as a source of anticancer drugs. Plant and microbes in ever changing environments. Plant science research and practices. (Edited by Shatya Shila Singh).

5. Vikas, Mehata, A.K., **Singh, C.,** Malik, A.K., Setia, A., Muthu, M.S. (**2024**). Alginate-Based Hydrogels as Drug Carriers. In: Jana, S. (eds) Biomaterial-based Hydrogels. Springer, Singapore. https://doi.org/10.1007/978-981-99-8826-6_2.

6. Saini, R., Mishra, A. K., **Singh, C**., Kumar, P., Shukla, P. K., Vishwakarma, J., & Tiwari, K. N. (2025). [Water stress in crop plants and its management: A metabolomic insight. In P. K. Srivastava, P. Parihar, & R. Upadhyay (Eds.), Water stress in crop plants and its management: [CRC Press], ISBN 9781032561233.

Edited Book/s

1.“Unlocking the secrets of extracellular vesicles and exosomes: a journey into the cutting-edge frontier of cellular communication” approved in Elsevier.

2. Antioxidant rich nutraceuticals: Harnessing Polyphenol for Health care” approved in Cambridge Scholar Publishing Ltd.

Patent/s

1. [Singh, C. "A Method for Biosynthesis of Zinc Glycinate Nano Complex and Composition Thereof" (Filed: 04/10/2023).

2. Singh, C. "Biosynthesis of Copper Glycinate Nano Complex Using Fruit Juice with Copper Sulphate" (Filed: 26/01/2024).
3. Singh C. Protein -enriched Millet-based cookies for malnourished, hypoproteinaemia and immunocompromised patients. (Filed: 4/02/2025). (Ref. no. TEMP/E-1/10800/2025-DEL).

Reviewer/Editor role

Reviewer in Scientific Reports, BMC complementary medicine and therapy and Vegetos.

Reference/s

1.Prof. KN Tiwari

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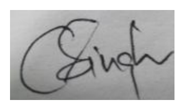
Declaration

I hereby declare that all information furnished above is true to the best of knowledge and belief and the documents in the support of information will be provided when required.

Date: 8/02/2025



(Chandrashekhar Singh)